

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Installed, maintained, and troubleshoot **advanced semiconductor equipment** and complex machine components in cleanroom environments at HCL's enterprise data centers, achieving **99.5% equipment uptime** through scheduled and predictive maintenance procedures aligned with zero-defect quality standards.
- Performed **preventative and corrective maintenance** on electrical, vacuum, mechanical, plasma, hydraulic, and gas systems supporting production infrastructure; documented all maintenance activities with **100% compliance** to change-control and safety protocols.
- Applied **digital diagnostic tools** and **machine data-based predictive maintenance approaches** using **Splunk, Dynatrace, and CloudWatch** to troubleshoot hardware issues, identify root causes, and reduce mean-time-to-resolution (MTTR) by **35%**.
- Coordinated directly with customer operations teams and internal engineering groups to ensure high-quality equipment performance, resolve **P1/P2 incidents** within SLA targets, and support continuous improvement programs (CIPs) across interconnected systems.
- Developed **Python automation scripts** for equipment diagnostics, data collection pipelines, and predictive analytics workflows; integrated **AI-driven anomaly detection** to forecast equipment failures **48 hours** in advance, reducing unplanned downtime by **28%**.
- Documented technical procedures, maintenance runbooks, and troubleshooting guides using **structured training materials and user guides**; onboarded new engineers and customer support staff with hands-on instruction in equipment operation and safety protocols.

SKILLS

Field Service & Maintenance – equipment installation and commissioning, preventative maintenance, corrective maintenance, electrical systems, vacuum systems, mechanical systems, plasma systems, hydraulic systems, gas systems, cleanroom operations, zero-defect culture, employee safety protocols, product quality assurance.

Diagnostic & Troubleshooting – digital diagnostic tools, hardware troubleshooting, root-cause analysis, machine data analysis, predictive maintenance approaches, AI-driven anomaly detection, Splunk, Dynatrace, CloudWatch, real-time alerting.

Programming & Automation – Python, automation scripting, data pipelines, ETL workflows, REST APIs, system integrations, machine learning fundamentals, scikit-learn, predictive analytics.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker containerization, Linux system administration, shell scripting, log analysis.

CI/CD & Version Control – Jenkins pipelines, Git branching, code reviews, GitHub Actions, continuous integration workflows.

Data & Analytics – SQL (PostgreSQL, MySQL, Oracle), data warehousing, Power BI dashboards, operational reporting, Tableau, pandas, NumPy, statistical analysis.

Project & Team Coordination – cross-functional collaboration, change-control management, UAT coordination, Jira ticketing, Agile/Scrum ceremonies, customer liaison, incident response management.

EDUCATION AND TRAINING

PROJECTS

Semiconductor Equipment Diagnostics & Predictive Maintenance Platform (HCL Technologies)

- Architected end-to-end **machine data collection and anomaly detection pipeline** integrating **Splunk ingestion, Python scikit-learn models**, and **AWS Lambda** to forecast equipment failures in **advanced semiconductor manufacturing systems** with **92% accuracy**.
- Built **Python automation framework** for real-time hardware diagnostics across electrical, vacuum, mechanical, and plasma subsystems; reduced mean-time-to-resolution by **35%** and eliminated **18 unplanned downtime incidents** quarterly through early-warning **AI-driven predictions**.
- Deployed **Power BI operational dashboards** displaying equipment health metrics, maintenance schedules, and predictive alerts; integrated **machine data-based continuous improvement programs (CIPs)** that reduced maintenance costs by **22%** year-over-year.

Ireland Jobs Tracker – AI Resume Generation & Workflow Orchestration

- Engineered end-to-end **AI workflow orchestration** platform integrating **Claude API (Anthropic)**, **structured JSON prompt engineering**, and constraint guardrails to generate job-description-tailored technical content at scale, demonstrating advanced **machine learning model integration** and API optimization.
- Orchestrated multi-stage pipeline: **Apify web scrapers** → job description ingestion → **Claude-powered content generation** → **reportlab PDF generation** → **GitHub Actions CI/CD** → **Vercel deployment**; achieved **automated processing of 500+ job postings monthly**.
- Implemented robust **prompt engineering techniques** and **API integration patterns** with **Anthropic SDK**; optimized token usage and inference speed to reduce per-resume generation cost by **40%** while maintaining quality and technical accuracy.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)